Sales Performance Analysis Report

1. Introduction

This project analyzes an online retail dataset to derive meaningful business insights. The goal is to understand sales performance, customer behavior, and revenue trends, and to provide actionable recommendations for business growth.

2. Data Cleaning & Preparation

The dataset was cleaned by removing missing values in CustomerID, filtering out negative quantities (returns), and eliminating records with zero unit price. Data types were corrected, and a new column 'Revenue' was created using Quantity × UnitPrice.

3. Data Analysis

3.1 Revenue Trends Over Time

Revenue shows fluctuations over time with noticeable peaks during certain months, indicating seasonal demand patterns.

3.2 Top-Selling Products

A small group of products contributes significantly to total revenue, highlighting a Pareto distribution.

3.3 Revenue by Country

The majority of revenue is generated from the United Kingdom, with other countries contributing relatively smaller shares.

3.4 Top Customers

High-value customers play a major role in revenue generation, indicating the importance of customer retention strategies.

4. Dashboard Overview

A Power BI dashboard was created to visualize key performance indicators including total revenue, total orders, and total customers. Visualizations include revenue trends, top products, country-wise revenue, and top customers.

5. Key Insights

- Revenue is highly concentrated in a few products and customers.
- The UK market dominates overall sales.
- Sales show seasonal spikes.
- Customer retention is crucial for sustained growth.

6. Recommendations

- Focus on top-performing products to maximize revenue.
- Expand operations in high-performing regions.
- Implement loyalty programs for high-value customers.
- Plan inventory for peak sales periods.

7. Conclusion

This analysis provides a comprehensive understanding of business performance and offers data-driven strategies to improve sales and customer engagement.